

Five-Layer Defense-in-Depth

L5 Operations
Safe multisig · key
rotation · bug bounty

Outer layers must be
breached before reaching
on-chain contract logic

L4 Runtime
Tenderly · The Graph ·
WebSocket · anomaly
detection

L3 AI / ML
Chainlink Functions ·
model registry · SHAP
guard

L2 Application
EIP-712 sign-in · JWT ·
rate limits

L1 Smart contracts
OpenZeppelin v5 · UUPS
· TimeLock · RBAC ·
ReentrancyGuard